

Submission Summary

Conference Name

International Conference on Computation, Communication, AND Intelligent Systems in Engineering

Track Name

Track 01: Computational Intelligence & Advanced Data Systems

Paper ID

519

Paper Title

Sales Orbis AI-Based Sales Pipeline Intelligence and Deal Win Prediction Platform

Abstract

The current study is concerned with the problem of estimating the probabilities of awarding contracts to suppliers in competitive procurement, where there exist several suppliers who compete on one particular project, and the relevant information comes gradually during the procurement process. Most of the existing machine learning approaches consider suppliers separately, and, as a result, give probabilities that are incoherent when interpreted together in the context of the project. In this study, we suggest a project-level formulation of predicting supplier awards, which results in coherent probabilities in case of stage-based missingness of information. Specifically, we consider the suppliers to be members of one choice set and include the possibility of non-awarding the contract, so that the probabilities obtained constitute a legitimate distribution over the project. Several baseline methods that consist of separate classification models followed by probability normalization are compared with the suggested approach. In addition, we study the issue of probability calibration, stage-related missingness, and behaviour of the models in case of absence of a particular procurement stage in the training dataset. To reduce the optimistic estimation caused by correlation among records of suppliers, we use projectlevel separation and project-clustered uncertainty quantification. Therefore, our evaluation concerns supplier-award prediction in terms of ranking and discrimination as well as in terms of coherence and calibration of probabilities, their robustness to missingness, and generalizability to procurement stages.

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Primary Subject Area

AI, Machine Learning & Deep Learning

Submission Files

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